

EV Ranking Method – Version 3 (Detailed + Matrix + Code)

1. Concept Summary

Teams are ranked using a Markov Chain where transition probabilities depend on past results. The steady-state distribution gives the EV score.

2. Initial Setup (1–1–0)

All teams start equally with 1–1–0 against each other to avoid division by zero.

3. Example Transition Matrix (India row)

From India →	India	England	Australia	SA
Probability	0.50	0.167	0.167	0.167

This row means: from India, there is 50% chance of staying and equal chance of moving to others.

4. EV Equation

We solve: $\pi P = \pi$

5. New Team (0–0–0)

A new team has no data, so we assign 1–1–0 vs all teams to keep system stable.

6. Python Example

```
import numpy as np

P = np.array([
    [0.5, 0.167, 0.167, 0.167],
    [0.167, 0.5, 0.167, 0.167],
    [0.167, 0.167, 0.5, 0.167],
    [0.167, 0.167, 0.167, 0.5]
])

# power iteration
pi = np.array([0.25, 0.25, 0.25, 0.25])
for _ in range(100):
    pi = pi @ P

print(pi)
```

This converges to equal probabilities at Day 0. After matches, P changes and rankings evolve.